

Pimpri Chinchwad Education Trust's
Pimpri Chinchwad College of Engineering (PCCoE)

(An Autonomous Institute)

Affiliated to Savitribai Phule Pune University (SPPU)

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TY B.Tech

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Semster: 5

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Assignment 5

Title: Create a simple functional college website/ travel agency website/insurance website using Node.js and Express, using any NoSQL dataset like MongoDB if required

Objectives:

- To learn basics of full stack development using Node.js and Express.js.
- To create a simple functional website (college/travel/insurance).
- To understand routing and request handling in Express.
- To build and connect frontend with backend.

Outcomes:

- Developed a working web application using Node.js.
- Understood routing and middleware in Express.
- Learned to handle client-server communication.
- Gained practical experience in full stack development.

Theory Concept:

Node.js and Express.js:

Node.js is a runtime environment that allows JavaScript to run on the server side. It uses an event-driven, non-blocking architecture which makes it efficient and scalable. Express.js is a web framework built on Node.js that simplifies server creation, routing, and handling HTTP requests and responses.

Full Stack Development:

Full stack development involves both frontend (HTML, CSS, JavaScript) and

backend (server, database, APIs). In this project, the frontend interacts with the backend server built using Node.js and Express.

Routing and Middleware:

Routing in Express defines how the application responds to client requests for different URLs such as home, about, or contact pages. Middleware functions are used to process requests before sending responses, such as parsing form data using body-parser.

MongoDB:

MongoDB is a NoSQL database that stores data in a flexible, JSON-like format called documents. It does not require a fixed schema, making it suitable for dynamic applications.

MongoDB Atlas:

MongoDB Atlas is a cloud-based database service that allows users to create, manage, and access MongoDB databases online. It eliminates the need for local database setup and provides features like scalability, security, and automatic backups.

MongoDB Connectivity using Atlas:

To connect a Node.js application with MongoDB Atlas, the following steps are used:

1. Create an account on MongoDB Atlas and set up a cluster.
2. Create a database user and configure network access (IP whitelist).
3. Obtain the connection string (URI) from Atlas.
4. Install MongoDB driver or Mongoose in the Node.js project.
5. Use the connection string in the application to establish a connection.

Example:

```
const mongoose = require('mongoose');

mongoose.connect('your_mongodb_connection_string', {
  useNewUrlParser: true,
  useUnifiedTopology: true
})
.then(() => console.log("Connected to MongoDB Atlas"))
.catch(err => console.log(err));
```

This connection allows storing and retrieving data from the cloud database.

Dataset Used:

None (Static or hardcoded data used)

Code:

Index.html

```
<%- include('partials/header') %>

<section class="hero-section">
  <div class="container hero-content">
    <h1 class="hero-title">Welcome to Aura Travel</h1>
    <p class="hero-subtitle">Find your next vacation destination here.</p>
    <a href="/packages" class="btn btn-primary hero-btn">Explore Destinations</a>
  </div>
</section>

<section class="featured-packages container">
  <div class="section-heading">
    <h2>Featured Experiences</h2>
  </div>

  <div class="packages-grid">
    <% packages.forEach(pkg => { %>
      <div class="package-card">
        <div class="card-img-wrapper">
          ">
        </div>
        <div class="card-body">
          <div class="price-tag">$<%= pkg.price %></div>
          <h3><%= pkg.title %></h3>
          <div class="meta-data">
            <span class="duration">Duration: <%= pkg.duration %></span>
          </div>
          <p class="card-desc"><%= pkg.description %></p>
          <a href="/book/<%= pkg._id %>" class="btn btn-outline">Book Now</a>
        </div>
      </div>
    <% }) %>
  </div>

  <div class="center-btn mt-4 text-center">
    <a href="/packages" class="btn btn-secondary">View All Packages</a>
  </div>
</section>

<%- include('partials/footer') %>
```

packages:

```
<%- include('partials/header') %>
```

```
<div class="page-header">
```

```
  <div class="container">
```

```
    <h1 class="page-title text-center">All Destinations</h1>
```

```
    <p class="page-subtitle text-center">Find your perfect escape from our curated  
collection.</p>
```

```
  </div>
```

```
</div>
```

```
<section class="packages-list container">
```

```
  <div class="packages-grid">
```

```
    <% packages.forEach(pkg => { %>
```

```
      <div class="package-card">
```

```
        <div class="card-img-wrapper">
```

```
          ">
```

```
        </div>
```

```
        <div class="card-body">
```

```
          <div class="price-tag">$<%= pkg.price %></div>
```

```
          <h3><%= pkg.title %></h3>
```

```
          <div class="meta-data">
```

```
            <span class="duration">Duration: <%= pkg.duration %></span>
```

```
          </div>
```

```
          <div>
```

```
            <strong>Destinations: </strong>
```

```
            <% pkg.destinations.forEach(dest => { %>
```

```
              <span><%= dest %> </span>
```

```
            <% }) %>
```

```
          </div>
```

```
          <p class="card-desc"><%= pkg.description %></p>
```

```
          <a href="/book/<%= pkg._id %>" class="btn btn-primary">Reserve Your
```

```
Spot</a>
```

```
        </div>
```

```
      </div>
```

```
    <% }) %>
```

```
  </div>
```

```
</section>
```

```
<%- include('partials/footer') %>
```

Book:

```
<%- include('partials/header') %>
```

```
<div class="page-header">
```

```
  <div class="container container-small">
```

```
    <h1 class="page-title text-center">Complete Booking</h1>
```

```
  </div>
```

```
</div>
```

```
<section class="booking-section container container-small">
```

```
  <div class="booking-card">
```

```
    <div class="booking-summary">
```

```
      <div class="summary-img">
```

```
        ">
```

```
      </div>
```

```
      <div class="summary-details">
```

```
        <h2><%= pkg.title %></h2>
```

```
        <div class="price-info">
```

```
          <span class="label">Price per person:</span>
```

```
          <span class="value">$<%= pkg.price %></span>
```

```
        </div>
```

```
        <div class="duration-info">
```

```
          <span class="label">Duration:</span>
```

```
          <span class="value"><%= pkg.duration %></span>
```

```
        </div>
```

```
      </div>
```

```
    </div>
```

```
    <form action="/book/<%= pkg._id %>" method="POST" class="booking-form">
```

```
      <div class="form-group">
```

```
        <label for="customerName">Full Name</label>
```

```
        <input type="text" id="customerName" name="customerName" class="form-control" required placeholder="John Doe">
```

```
      </div>
```

```
      <div class="form-group">
```

```
        <label for="email">Email Address</label>
```

```
        <input type="email" id="email" name="email" class="form-control" required placeholder="john@example.com">
```

```
      </div>
```

```
      <div class="form-group">
```

```
        <label for="phone">Phone Number</label>
```

```
        <input type="tel" id="phone" name="phone" class="form-control" required placeholder="+1 234 567 8900">
```

```
      </div>
```

```

    <div class="form-group">
      <label for="travelDate">Select Travel Date</label>
      <input type="date" id="travelDate" name="travelDate" class="form-control"
required>
    </div>

    <div class="form-group">
      <label for="numberOfTravelers">Number of Travelers</label>
      <input type="number" id="numberOfTravelers" name="numberOfTravelers"
class="form-control" min="1" value="1" required>
    </div>

    <div class="total-bar">
      <span>Total Expected:</span>
      <span class="total-price" id="totalPriceDisplay">$<%= pkg.price %></span>
    </div>

    <button type="submit" class="btn btn-primary mt-4">Confirm Booking</button>
  </form>
</div>
</section>

<script>
  // Simple script to update total price dynamically
  const pricePerPerson = <%= pkg.price %>;
  const inputTravelers = document.getElementById('numberOfTravelers');
  const displayTotal = document.getElementById('totalPriceDisplay');

  inputTravelers.addEventListener('input', (e) => {
    const travelers = parseInt(e.target.value) || 0;
    const total = pricePerPerson * travelers;
    displayTotal.innerText = '$' + total.toLocaleString();
  });
</script>

<%- include('partials/footer') %>

```

App.js

```

require('dotenv').config();
const express = require('express');
const mongoose = require('mongoose');
const path = require('path');
const bodyParser = require('body-parser');

const app = express();
const PORT = process.env.PORT || 3000;

```

```

const MONGODB_URI = process.env.MONGODB_URI ||
'mongodb://127.0.0.1:27017/travelagency';

// Models
const Package = require('./models/Package');
const Booking = require('./models/Booking');

// Middleware Setup
app.set('view engine', 'ejs');
app.set('views', path.join(__dirname, 'views'));
app.use(express.static(path.join(__dirname, 'public')));
app.use(bodyParser.urlencoded({ extended: true }));
app.use(bodyParser.json());

// Database Connection
mongoose.connect(MONGODB_URI)
.then(async () => {
  console.log(`Connected to MongoDB at ${MONGODB_URI}`);
  await seedDatabase();
}).catch(err => {
  console.error('Error connecting to MongoDB. Please ensure MongoDB is running globally
or start the service.', err);
});

// Seed Data
async function seedDatabase() {
  try {
    const count = await Package.countDocuments();
    if (count === 0) {
      console.log("Seeding database with initial travel packages...");
      const initialPackages = [
        {
          title: "Maldives Paradise Escape",
          description: "Experience the crystal clear waters and overwater bungalows of
the Maldives. Perfect for couples and honeymooners.",
          price: 2499,
          image: "https://images.unsplash.com/photo-1514282401047-
d79a71a590e8?ixlib=rb-4.0.3&auto=format&fit=crop&w=800&q=80",
          duration: "7 Days / 6 Nights",
          destinations: ["Male", "Baa Atoll"]
        },
        {
          title: "Swiss Alps Adventure",
          description: "Breathtaking mountain views, scenic train rides, and pure alpine
air. An unforgettable journey through Switzerland.",
          price: 1899,
          image: "https://images.unsplash.com/photo-1530122037265-
a5f1f91d3b99?ixlib=rb-4.0.3&auto=format&fit=crop&w=800&q=80",

```

```

        duration: "5 Days / 4 Nights",
        destinations: ["Zurich", "Interlaken", "Zermatt"]
      },
      {
        title: "Bali Cultural Immersion",
        description: "Discover ancient temples, lush rice terraces, and beautiful beaches
in the Island of the Gods.",
        price: 1299,
        image: "https://images.unsplash.com/photo-1537996194471-
e657df975ab4?ixlib=rb-4.0.3&auto=format&fit=crop&w=800&q=80",
        duration: "10 Days / 9 Nights",
        destinations: ["Ubud", "Seminyak", "Nusa Penida"]
      },
      {
        title: "Kyoto Heritage Tour",
        description: "Walk through history in Japan's cultural heart. Experience tea
ceremonies, bamboo forests, and stunning shrines.",
        price: 2199,
        image: "https://images.unsplash.com/photo-1493976040374-
85c8e12f0c0e?ixlib=rb-4.0.3&auto=format&fit=crop&w=800&q=80",
        duration: "8 Days / 7 Nights",
        destinations: ["Kyoto", "Osaka", "Nara"]
      }
    ];
    await Package.insertMany(initialPackages);
    console.log("Successfully seeded travel packages!");
  }
} catch (err) {
  console.error("Error seeding database", err);
}
}

// Routes
app.get('/', async (req, res) => {
  try {
    const packages = await Package.find().limit(3);
    res.render('index', { packages });
  } catch (err) {
    res.status(500).send("Database error occurred.");
  }
});

app.get('/packages', async (req, res) => {
  try {
    const packages = await Package.find();
    res.render('packages', { packages });
  } catch (err) {
    res.status(500).send("Database error occurred.");
  }
});

```



```

    }
  });

  app.get('/book/:id', async (req, res) => {
    try {
      const pkg = await Package.findById(req.params.id);
      if (!pkg) return res.status(404).send('Package not found');
      res.render('book', { pkg });
    } catch (err) {
      res.status(500).send("Database error occurred.");
    }
  });

  app.post('/book/:id', async (req, res) => {
    try {
      const pkg = await Package.findById(req.params.id);
      if (!pkg) return res.status(404).send('Package not found');

      const { customerName, email, phone, travelDate, numberOfTravelers } = req.body;

      const travelers = parseInt(numberOfTravelers, 10);
      const totalPrice = pkg.price * travelers;

      const booking = new Booking({
        packageId: pkg._id,
        customerName,
        email,
        phone,
        travelDate,
        numberOfTravelers: travelers,
        totalPrice
      });

      await booking.save();
      res.render('success', { booking, pkg });

    } catch (err) {
      console.error(err);
      res.status(500).send("Error creating booking.");
    }
  });

  // Start Server
  app.listen(PORT, () => {
    console.log(`Travel Agency Server is running on http://localhost:${PORT}`);
  });

```

Welcome to Aura Travel

Find your next vacation destination here.

Explore Destinations

Featured Experiences



All Destinations

Find your perfect escape from our curated collection.



\$2499

Maldives Paradise Escape

Duration: 7 Days / 6 Nights

Destinations: Male Baa Atoll
Experience the crystal clear waters and overwater bungalows of the Maldives. Perfect for couples and honeymooners.

Reserve Your Spot



\$1899

Swiss Alps Adventure

Duration: 5 Days / 4 Nights

Destinations: Zurich Interlaken Zermatt
Breathtaking mountain views, scenic train rides, and pure alpine air. An unforgettable journey through Switzerland.

Reserve Your Spot



\$1299

Bali Cultural Immersion

Duration: 10 Days / 9 Nights

Destinations: Ubud Seminyak Nusa Penida
Discover ancient temples, lush rice terraces, and beautiful beaches in the Island of the Gods.

Reserve Your Spot

Complete Booking



Swiss Alps Adventure

Price per person: \$1899
Duration: 5 Days / 4 Nights

Full Name

John Doe

Email Address

john@example.com

Phone Number

+1 234 567 8900

Select Travel Date

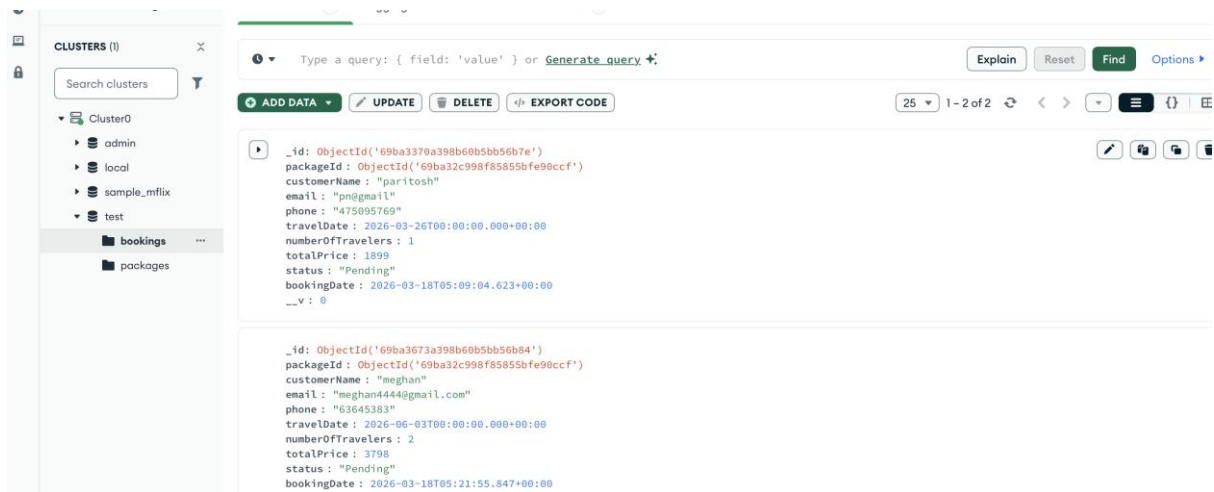
20/03/2026

Number of Travelers

2

Total Expected: \$3,798

Connection with MongoDB:



Conclusion

This practical provided hands-on experience in full stack development using Node.js and Express. It also helped in understanding how to connect applications with MongoDB Atlas for cloud-based data storage.